

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 2: Nutrition in Animals

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	2 — Nutrition in Animals
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. These are the hardest traps: deciding which single step a multi-event scenario isolates, separating WHERE enzymes sit (gut cavity vs cells vs blood) from WHAT they do, distinguishing physical emulsification from chemical breakdown, tracing a failure to the exact step it breaks while proving the earlier steps still ran, untangling rumen/reticulum/omasum/abomasum and cud/chyme, reasoning quantitatively about teeth and surface area, and mapping Amoeba's single-cell, temporary-structure plan onto a human's many fixed organs without over- or under-counting the five steps. Do not write on this paper — mark answers on your answer sheet.

1. A researcher tags a single fat molecule in a meal and tracks it. It is bitten off and swallowed, broken by an enzyme into smaller pieces, those pieces cross the gut wall into a transport fluid, and finally a cell uses them to build part of its own boundary. The researcher must label the FOUR distinct steps these four events belong to, in order. Which list is exactly right?

- (A) Ingestion, digestion, assimilation, absorption
- (B) Digestion, ingestion, absorption, egestion
- (C) Ingestion, absorption, digestion, assimilation
- (D) Ingestion, digestion, absorption, assimilation

2. A patient takes food in normally, breaks it down normally, and the simple molecules cross into the blood normally, but a defect leaves those molecules circulating unused so the patient weakens over weeks. To prove which single step has failed, which observation is logically decisive?

- (A) Food enters the mouth, which shows ingestion is the failing step
- (B) Simple molecules are present in the blood yet the body wastes away, which shows ingestion, digestion and absorption all succeeded and only the final using-by-cells step is broken
- (C) The molecules never reach the blood, which shows absorption is the failing step
- (D) Undigested waste is being kept inside, which shows egestion is the failing step

3. A teacher draws two exits from the body. Exit 1 carries out plant fibre and dead lining cells that travelled through the food canal without ever being broken into nutrients. Exit 2 carries out a gas formed when cells release energy from glucose. A student labels both 'egestion'. What single correction is needed?

- (A) Only Exit 2 is egestion; Exit 1 is excretion because it carries dead cells
- (B) Only Exit 1 is egestion (undigested food residue leaving the canal); Exit 2 is excretion, removing waste made inside cells
- (C) Both are egestion, since any waste leaving the body is egestion regardless of source
- (D) Both are excretion, since both remove unwanted matter from the body

4. An Amoeba, a python and a leaf-cutter scenario are filmed: the Amoeba wraps a particle, the python unhinges its jaw to swallow prey whole, and a butterfly uncoils a long tube to draw nectar. A judge insists all three clips, despite looking nothing alike, capture exactly one shared step. He is right. Which step, and what makes them the same despite the differences?

- (A) Digestion: each clip shows the food being chemically broken down as it enters
- (B) Ingestion: each is simply food being taken into the organism, so the wildly different methods do not change the step
- (C) Absorption: each clip shows nutrients crossing directly into body fluids at the feeding point
- (D) Assimilation: each clip shows the organism putting the food to use as it is taken in

5. A housefly releases digestive juice onto solid food OUTSIDE its body, then sucks up the resulting liquid. A student concludes the housefly therefore performs absorption before ingestion. Identify the precise flaw.

- (A) No absorption happens outside the body; the external juice only begins digestion, and the liquefied food is then taken in (ingestion) before being absorbed inside
- (B) The student is right: the juice absorbs nutrients into the fly outside the body, so absorption truly precedes ingestion
- (C) The housefly skips digestion entirely, since it only sucks liquid that needs no breaking down
- (D) The housefly skips ingestion, since the food is already liquid when it reaches the fly

6. A biologist remarks that an Amoeba shows an efficiency a human stomach lacks: the same temporary extension it forms to creep toward food is also the extension it uses to capture that food. What general body-plan point does this illustrate?

- (A) The extension digests and absorbs the food at the same time, doing two chemical jobs at once
- (B) The extension is a permanent organ, just smaller than a human's, dedicated only to feeding
- (C) The extension makes the Amoeba's own bile, letting it digest fat without any liver
- (D) Being a single cell, the Amoeba improvises one temporary structure that serves two jobs (moving and feeding), instead of maintaining a separate fixed organ for each task as a multi-organ body does

7. An adult set has 8 incisors, 4 canines, 8 premolars and 12 molars. A claim states 'the cutting-and-tearing teeth outnumber the grinding teeth.' Evaluate the claim with the numbers.

- (A) True: cutting-and-tearing teeth total 20 and grinding teeth total 12
- (B) False: cutting-and-tearing teeth total 12 while grinding teeth total 20, so grinding teeth are the larger group
- (C) True: there are 16 of each kind, so neither outnumbers the other
- (D) False, but only because all 32 teeth can grind

8. A dentist's drill enters a tooth and reports: a very hard outer shell, then a softer middle layer occupying most of the tooth, then a central space where the patient suddenly feels sharp pain. Name the three regions in the order the drill met them.

- (A) Dentine, then enamel, then the pulp cavity
- (B) Enamel, then the pulp cavity, then dentine
- (C) The pulp cavity, then dentine, then enamel
- (D) Enamel, then dentine, then the pulp cavity

9. Cavity X has eaten deep into the dentine yet the patient feels no pain. Cavity Y is shallower but causes sharp pain. A student says this proves the dentine is harder than the enamel. What actually reconciles the two facts?

- (A) Dentine is indeed the hardest layer, so the deep Cavity X meets resistance and feels nothing
- (B) Cavity X is painless because the nerves are stored in the enamel, which it bypassed
- (C) Both descriptions cannot be true, because a deeper cavity must always hurt more
- (D) Pain depends on reaching the nerves in the pulp, not on depth: Cavity X is deep in dentine but has not touched the pulp, while shallow Cavity Y has reached or nearly reached the pulp

10. Tooth decay runs: leftover sugar on teeth, then mouth bacteria feed on it, then bacteria release acid, then acid dissolves enamel. A child cannot brush after lunch and wants the action that breaks this chain at the EARLIEST possible link. Which works earliest?

- (A) Swallowing an antacid so stomach acid does not rise and reach the teeth
- (B) Rinsing the mouth with water to wash away the leftover sugar before bacteria can feed on it

- (C) Eating a sweet dessert so the bacteria become full and stop releasing acid
- (D) Chewing tough food so the enamel hardens and resists the acid

11. Two grass-eaters and one flesh-eater leave only their skulls. Skull P has many broad, ridged back teeth and small front teeth; skull Q has prominent pointed teeth flanking the front teeth and fewer flat ones. From teeth alone, what is the safest conclusion?

- (A) Skull P likely ate plants (broad ridged teeth grind) and skull Q likely ate flesh (prominent pointed teeth tear); tooth form reflects diet
- (B) Skull Q belonged to the larger animal, since pointed teeth only grow in big animals
- (C) Both skulls have identical teeth and only their jaw muscles differed
- (D) Tooth shape is random, so nothing reliable about diet can be concluded

12. A person loses all premolars and molars but keeps every incisor and canine. Of these everyday tasks, which becomes MOST difficult, based purely on what each tooth type does?

- (A) Biting a piece off an apple, because the cutting teeth were lost
- (B) Tearing meat from a bone, because the tearing teeth were lost
- (C) Grinding tough fibrous food into a fine pulp, because the grinding teeth are the ones that were lost
- (D) Swallowing liquids, because teeth are essential for swallowing

13. A student chews a starchy cracker for two minutes and it turns sweet; chewing a piece of cheese (mostly fat and protein) for the same time gives no sweetness. What does this contrast prove about the enzyme in saliva?

- (A) Saliva digests every food equally, so the cheese must have secretly turned sweet too
- (B) Saliva acts only on protein, which is why neither food became sweet
- (C) Saliva acts on fat, so the fatty cheese should have tasted sweeter than the cracker
- (D) Saliva's enzyme acts on starch, turning it into sugar, but does not break down fat or protein, so the cheese stays unsweet

14. Two warm tubes of starch are prepared: Tube 1 receives fresh saliva, Tube 2 receives saliva that was boiled and then cooled. After thirty minutes only Tube 1 shows sugar. Boiling is known to destroy enzymes. What is the single strongest conclusion?

- (A) Warmth by itself turned the starch into sugar, since both tubes were warm
- (B) Fresh saliva contains an enzyme that breaks starch into sugar, and boiling destroyed that enzyme in Tube 2
- (C) Saliva is acidic, and the acid (not an enzyme) broke the starch in Tube 1
- (D) Boiled saliva is more powerful, so Tube 2 should have shown more sugar

15. Saliva offers a dry biscuit two distinct helps. A student names only 'it begins digesting the starch.' Which second help is overlooked, and why does swallowing depend on it?

- (A) Saliva moistens and softens the dry biscuit into a slippery lump, without which it could not slide easily down the food pipe
- (B) Saliva absorbs the biscuit's sugar into the blood in the mouth, easing the load downstream
- (C) Saliva produces hydrochloric acid in the mouth to soften the biscuit before swallowing
- (D) Saliva cools the biscuit so it does not burn the food pipe on the way down

16. A person swallows water while hanging fully upside-down and it still reaches the stomach. A friend insists 'this only works because some water trickles down by gravity.' Which single point most decisively refutes the friend?

- (A) Here gravity is pulling the water away from the stomach, so the water reaching it must be pushed by muscular waves of the food pipe, not pulled down
- (B) Gravity still helps, because water is heavy and always finds its way to the stomach
- (C) The food pipe is a straight vertical drop, so the body's position does not matter
- (D) The food pipe absorbs the water directly, so it never needs to travel to the stomach

17. A medicine freezes the food-pipe muscles so they can no longer make their squeezing waves, while the mouth, stomach, intestines and saliva all keep working. A person chews and swallows a sandwich. What is the most direct problem?

- (A) The stomach stops churning, so the sandwich rots inside the food pipe
- (B) The swallowed sandwich cannot be pushed down the food pipe to the stomach, so it lodges there
- (C) Salivary digestion stops, so the sandwich turns bitter in the mouth
- (D) Water is over-absorbed in the food pipe, drying the sandwich into a hard lump

18. Three sites are ranked by how long food typically stays: a wave-pushing tube it passes in seconds, a churning bag it stays in for a few hours, and a long coiled tube it stays in for many hours. Name them from shortest stay to longest stay.

- (A) Stomach (seconds), oesophagus (a few hours), small intestine (many hours)
- (B) Oesophagus (seconds), stomach (a few hours), small intestine (many hours)
- (C) Oesophagus (seconds), small intestine (a few hours), stomach (many hours)
- (D) Small intestine (seconds), oesophagus (a few hours), stomach (many hours)

19. Gastric juice has hydrochloric acid, mucus and pepsin. A student assigns each a job and gets exactly one wrong. Which assignment is the error?

- (A) 'Mucus digests the proteins' is the error; mucus protects the wall, while pepsin is the protein-digesting enzyme
- (B) 'Hydrochloric acid kills germs in the food' is the error
- (C) 'Pepsin begins digesting proteins' is the error
- (D) 'Hydrochloric acid makes the surroundings acidic for pepsin' is the error

20. A patient makes almost no hydrochloric acid in the stomach. Two harmful effects are expected, each traceable to a normal job of the acid. Which pair is correct?

- (A) Starch digestion fails everywhere, and bile can no longer reach the small intestine
- (B) Fat can no longer be emulsified, and water absorption in the large intestine stops
- (C) Pepsin works poorly so stomach protein digestion is impaired, and swallowed germs are less effectively killed
- (D) Saliva stops being made, and the food pipe loses its peristalsis

21. The stomach wall is itself built mostly of protein and is bathed in protein-digesting pepsin and acid, yet it is not digested. A student says 'the wall must contain no protein for pepsin to find.' Why is that wrong, and what is the real protection?

- (A) The student is right; the stomach wall genuinely contains no protein
- (B) Bile from the liver constantly neutralises the stomach acid and protects the wall
- (C) The wall is indeed protein; it is protected because a layer of mucus lines the inner surface and shields it from the acid and pepsin
- (D) The wall is made of bone, which pepsin cannot digest

22. Pepsin needs strongly acidic surroundings, but the enzymes waiting in the small intestine do not. Tracing the half-digested protein paste as it leaves the stomach into the small intestine, what must happen for digestion to continue smoothly?

- (A) It must be made even more acidic, since all digestive enzymes prefer strong acid
- (B) Its strong acidity must be reduced (neutralised) in the small intestine, because the intestinal enzymes work in non-acidic conditions
- (C) It must be kept exactly as acidic as the stomach so pepsin keeps working in the intestine
- (D) It must be dried out completely before the intestinal enzymes can act

23. Two semi-processed materials need labels: (i) the thick semi-liquid paste the human stomach pushes into the small intestine, and (ii) the partly digested grass a cow brings back up to chew. A student labels (i) 'cud' and (ii) 'chyme'. Correct the labels.

- (A) Both should be called chyme, since both are partly digested food
- (B) Both should be called cud, since cud covers any partly digested mass
- (C) (i) is cud and (ii) is bile, since a cow re-chews bile
- (D) (i) is chyme and (ii) is cud; the student swapped the two names

24. Two model guts have equal length and equal transit time. Gut A has a smooth lining; Gut B's lining is packed with millions of tiny finger-like projections. Gut B absorbs far more. The single best reason the projections help is that they:

- (A) Greatly increase the inner surface area, packing far more nutrient-absorbing contact into the same length
- (B) Speed the food up so it wastes less time in the gut

(C) Make Gut B more acidic, which dissolves nutrients faster

(D) Produce extra bile inside Gut B to break down the fats

25. A disease flattens the villi of the small intestine until the lining is smooth, yet the person keeps eating and digesting normally. Trace the consequence over months to body weight.

(A) Mouth digestion speeds up to compensate, so the person gains weight

(B) The stomach churns harder to force nutrients through, so weight stays the same

(C) Absorption falls sharply because the surface area is lost, so the person grows undernourished and loses weight despite eating

(D) The liver makes extra bile to absorb the nutrients instead, so nothing changes

26. Inside each villus run tiny blood vessels. A rice meal is digested to glucose, the glucose crosses the villus wall into those vessels, and is then carried to a working muscle that uses it. Name, in order, the step happening AT the villus and the step happening AT the muscle.

(A) Digestion at the villus, then absorption at the muscle

(B) Absorption at the villus, then assimilation at the muscle

(C) Assimilation at the villus, then absorption at the muscle

(D) Absorption at the villus, then egestion at the muscle

27. A student claims 'the villi must be where digestion is completed, because that is where the simple molecules finally appear.' Correct the reasoning precisely.

(A) The simple molecules appear at the villi only because they are absorbed there; the digestion that made them happened earlier in the gut cavity before reaching the villi

(B) The student is right; the villi are the body's main digesting organs

(C) The villi both digest and store the molecules, which is why they appear there

(D) The molecules appear at the villi because the villi churn the food like a small stomach

28. Compare two events in the small intestine: a fat globule being broken into many tiny droplets, and an amino acid crossing the wall into the blood. A student calls both 'digestion.' Give the correct labels.

(A) Both are absorption, since both move material across the intestinal wall

(B) Both are digestion, since both involve molecules changing in the intestine

(C) Breaking the fat into droplets is part of digestion (physically preparing fat for enzymes), but the amino acid crossing into the blood is absorption, not digestion

(D) Breaking the fat into droplets is absorption, and the amino acid crossing is digestion

29. A student insists 'the longest organ always does the most important nutrient work, so the long small intestine, being longer than the large intestine, must also be where water is recovered.' Pinpoint the error.

- (A) The student is right; the longer small intestine recovers the water as well as absorbing nutrients
- (B) The error is only about importance; the small intestine still recovers most of the water
- (C) Length does not assign jobs: the small intestine mainly absorbs nutrients, while recovering water from the residue is the large intestine's distinct job
- (D) Both organs recover water equally, so neither is specially responsible

30. A patient's large intestine reabsorbs far too LITTLE water from the residue. Predict the symptom and contrast it with reabsorbing too much.

- (A) Too little water gives hard, dry stools, while too much gives watery stools
- (B) Too little water causes vomiting of stomach acid, while too much speeds fat digestion
- (C) Either way the stool is unchanged, because the large intestine does not handle water
- (D) Too little water gives loose, watery stools, while reabsorbing too much gives hard, dry stools (constipation)

31. A child swallows a small, smooth, indigestible plastic bead. Following the five steps, predict its fate and name the only steps it can take part in.

- (A) It is digested by stomach acid into harmless nutrients and then absorbed
- (B) It is assimilated into the child's bones along with calcium
- (C) It is excreted as a chemical waste made inside the child's cells
- (D) It is taken in (ingestion) and later passed out with the food residue (egestion); it cannot be digested, absorbed or assimilated

32. The liver is the body's largest gland and makes bile, yet bile contains NO digestive enzyme. A student concludes bile cannot help digestion at all. Refute this and state bile's real action and where it occurs.

- (A) Bile, made by the liver and acting in the small intestine, breaks large fat globules into tiny droplets so fat-digesting enzymes can act on them more easily
- (B) The student is right; with no enzyme, bile is useless for digestion
- (C) Bile chemically cuts fat into amino acids in the stomach without any enzyme
- (D) Bile turns fat into starch in the small intestine so starch enzymes finish it

33. A patient's gall bladder is removed. Over the next year, fatty meals trouble them most, while starchy and protein meals are handled almost normally. What does this pattern reveal about what the gall bladder stored and what that fluid mainly assists?

- (A) The gall bladder stored bile, and bile mainly assists the digestion of fats, so only fatty meals are noticeably affected
- (B) The gall bladder made pepsin, which mainly digests fats, so fatty meals suffer

- (C) The gall bladder stored amylase, so its loss should have harmed starchy meals instead
- (D) The gall bladder produced stomach acid, so all three meal types should suffer equally

34. A bile-duct blockage stops bile reaching the small intestine, while saliva, gastric juice and pancreatic enzymes all still arrive. Of starch, protein and fat, which is MOST hampered, and what happens to the others?

- (A) Fat is most hampered because, without bile, it stays in large globules that enzymes struggle to act on; starch and protein digestion continue almost normally
- (B) Starch is most hampered, because bile is the main starch-digesting enzyme
- (C) Protein is most hampered, because bile activates the protein enzymes in the intestine
- (D) All three are hampered equally, because bile is required to digest every food type

35. Three secretions are matched to one food each, in both the food acted on AND the nature of the action. Which matching is fully correct?

- (A) Saliva chemically starts protein; gastric pepsin starts fat; bile chemically breaks starch
- (B) Saliva emulsifies fat; gastric pepsin starts starch; bile chemically breaks protein
- (C) All three chemically break the same food, starch, in different organs
- (D) Saliva chemically starts starch; gastric pepsin chemically starts protein; bile physically emulsifies fat without chemically breaking it

36. Pancreatic juice acts on carbohydrates, proteins AND fats. If a tube carrying ONLY pancreatic juice were blocked while saliva and gastric juice still flowed, which broad outcome best fits?

- (A) Only starch digestion in the mouth fails, since the pancreas makes the salivary enzyme
- (B) Digestion of carbohydrates, proteins and fats in the small intestine is all impaired together, because the pancreas supplies enzymes for all three there
- (C) Only protein digestion in the stomach fails, since the pancreas makes pepsin
- (D) Only water absorption in the large intestine fails, since the pancreas absorbs water

37. Why must the strongly acidic chyme leaving the stomach be neutralised before the pancreatic enzymes act on it in the small intestine?

- (A) Because the pancreatic enzymes need even stronger acid than pepsin to work
- (B) Because neutralising the acid is itself what digests the carbohydrates and fats
- (C) Because the pancreatic enzymes work in non-acidic conditions, so the stomach's acid must first be neutralised for them to function
- (D) Because the small intestine cannot hold any liquid that is not perfectly dry

38. A ruminant's four-chambered stomach is studied. Microbes first attack cellulose in the largest chamber; later, acid-and-enzyme digestion like a human stomach occurs in a different chamber. Name these two chambers in that order, avoiding the common swap.

- (A) Microbial cellulose breakdown in the abomasum; acid-and-enzyme digestion in the rumen
- (B) Microbial cellulose breakdown in the rumen; acid-and-enzyme digestion in the reticulum
- (C) Microbial cellulose breakdown in the omasum; acid-and-enzyme digestion in the rumen
- (D) Microbial cellulose breakdown in the rumen; human-like acid-and-enzyme digestion in the abomasum

39. A cow and a person are each fed only fresh grass for a month. The cow thrives; the person grows weak. A student says 'the cow simply has a bigger stomach.' What is the deeper, correct reason?

- (A) The cow has rumen microbes plus cud re-chewing that together break down the cellulose in grass; the person has neither and cannot digest cellulose
- (B) The cow's stomach is merely larger, so it holds more grass and gets more from the same food
- (C) The person has no stomach, so cannot hold the grass long enough to digest it
- (D) Grass needs no digestion, so the cow simply eats more of it than the person does

40. A goat regurgitates grass, chews it long and finely, and re-swallows it. A student concludes the goat digests its cellulose entirely in the mouth, since that is where the long chewing happens. Correct the view.

- (A) The student is right; the long chewing fully digests the cellulose in the mouth
- (B) The cellulose is broken down in the food pipe as the cud is re-swallowed
- (C) The mouth only grinds the cud finer; the cellulose itself is broken down by microbes in the rumen, not in the mouth
- (D) The cellulose is never digested at all; the goat merely stores the cud forever

41. Re-chewing the cud is said to help a cow get more energy from grass. By what chain of cause and effect does finer grinding lead to more energy released?

- (A) Finer grinding turns the cellulose directly into sugar in the mouth, absorbed there
- (B) Finer grinding exposes more surface of the grass, so rumen microbes and enzymes can break the cellulose down more completely, freeing more usable nutrients
- (C) Finer grinding heats the grass, and the heat alone releases the energy in cellulose
- (D) Finer grinding turns the cellulose into protein, which carries more energy

42. An Amoeba captures a particle with pseudopodia, traps it in a structure, digests it there, absorbs the useful part, and later throws the rest out. Name the trapping structure and the step by which the leftover is thrown out.

- (A) The trapping structure is a gall bladder, and the throwing-out is excretion of cell-made waste
- (B) The trapping structure is a villus, and the throwing-out is absorption of the leftover
- (C) The trapping structure is a food vacuole, and the throwing-out of the leftover is egestion when the vacuole reaches the cell surface
- (D) The trapping structure is a rumen, and the throwing-out is assimilation of the leftover

43. In an Amoeba, digestive enzymes are released INTO the food vacuole, a closed pocket inside the cell. The closest human parallel for WHERE digestive enzymes act is:

- (A) Inside the blood, where enzymes act on absorbed nutrients after they enter
- (B) Inside the body's muscle cells, where nutrients are finally burnt for energy
- (C) On the outer skin surface, before the food is swallowed into the body
- (D) Inside the cavity (space) of the stomach and intestine, which are enclosed spaces, not inside the body's own cells or in the blood

44. A teacher claims: 'An Amoeba and a human both complete all five steps of nutrition; the difference is only in the body plan that carries them out.' Choose the statement that both supports the claim and correctly states the difference.

- (A) Only the human truly digests food; the Amoeba merely engulfs and stores it without digestion
- (B) The Amoeba absorbs nutrients but never egests, so it completes only four of the five steps
- (C) Both ingest, digest, absorb, assimilate and egest; the Amoeba does all five within one cell while the human spreads them across many specialised organs
- (D) The human ingests and digests but never assimilates, so neither truly completes all five

45. Trace a meal of buttered toast: it has starch (toast) and fat (butter). Name the organ where the starch's digestion FIRST begins and the organ where the butter's fat is MAINLY emulsified and chemically digested.

- (A) Starch digestion first begins in the mouth; the fat is mainly emulsified and digested in the small intestine
- (B) Both the starch and the fat first begin digestion in the stomach
- (C) Starch digestion first begins in the small intestine; the fat is digested in the mouth
- (D) Both the starch and the fat are digested only in the large intestine

46. Put the whole pathway together for a paneer (protein-rich) meal: name where its protein digestion BEGINS, where that digestion is COMPLETED, and where its building blocks are ABSORBED.

- (A) Begins in the mouth, completed in the stomach, absorbed in the large intestine
- (B) Begins in the stomach, completed in the small intestine, and the building blocks are absorbed in the small intestine

- (C) Begins in the small intestine, completed in the stomach, absorbed in the mouth
- (D) Begins in the large intestine, completed in the mouth, absorbed in the stomach

47. A meal of sweet, watery fruit is eaten. Comparing where its dissolved SUGAR is mainly absorbed with where its WATER is mainly absorbed, the correct pairing is:

- (A) Sugar is mainly absorbed in the small intestine; water is mainly absorbed in the large intestine
- (B) Both sugar and water are mainly absorbed in the stomach
- (C) Sugar is mainly absorbed in the large intestine; water in the small intestine
- (D) Both sugar and water are mainly absorbed in the mouth

48. Trace a single sip of milk (with fat, protein and milk-sugar) at the very moment it first enters the mouth. Exactly how many of the five steps have been completed for it, and which step is occurring?

- (A) One step (digestion) is completed and absorption is now occurring
- (B) Two steps (ingestion and digestion) are completed and assimilation is now occurring
- (C) Zero steps are completed; the step occurring at that moment is ingestion, the taking-in of the milk
- (D) Four steps are completed and only egestion remains

49. A student writes the five steps and swaps two so the list reads: ingestion, digestion, assimilation, absorption, egestion. Name the single error and why it cannot stand.

- (A) Egestion is in the wrong place; it should come before digestion
- (B) Ingestion should come last, since food is taken in only after it is digested
- (C) Assimilation is placed before absorption, which is impossible because nutrients must first enter the blood (absorption) before any cell can use them (assimilation)
- (D) There is no error; the two orders mean the same thing

50. A nutrition chart needs three glands labelled by description: the largest gland that makes bile, the small sac storing that bile between meals, and the gland whose juice acts on all three food types. In order, these are:

- | | |
|-----------------------------------|-----------------------------------|
| (A) Liver, pancreas, gall bladder | (B) Liver, gall bladder, pancreas |
| (C) Gall bladder, liver, pancreas | (D) Pancreas, gall bladder, liver |

51. A medicine completely stops peristalsis in the food pipe but leaves the mouth, stomach, intestines, saliva and all juices working. A different medicine instead stops the salivary enzyme only. Which single eating problem follows from the FIRST medicine?

- (A) Starch tastes less sweet on long chewing, because no starch is broken in the mouth
- (B) The stomach can no longer churn the food it receives
- (C) Swallowed food cannot be carried down the food pipe to the stomach, so it gets stuck on the way
- (D) Water can no longer be recovered from the residue in the large intestine

52. A snake unhinges its jaw to swallow prey whole while an Amoeba wraps a particle in pseudopodia. A student claims these differ in their step of DIGESTION. Pinpoint the conceptual mistake.

- (A) They differ in their method of INGESTION (how food is taken in), not digestion; digestion is the later chemical breakdown, which these scenes do not show differing
- (B) They differ in absorption, since each moves nutrients into its body fluids differently
- (C) The student is right; swallowing whole and engulfing are two kinds of digestion
- (D) They differ in assimilation, since each builds tissue from food in its own way

53. A person eats well and their gut digests and absorbs normally, but their cells cannot put the absorbed glucose to use, so they weaken. Now compare: in a DIFFERENT person, the gut absorbs nothing into the blood at all. Which steps are failing, respectively?

- (A) In the first person absorption is failing; in the second person digestion is failing
- (B) In both people the same step, digestion, is failing
- (C) In the first person assimilation is failing; in the second person absorption is failing
- (D) In the first person egestion is failing; in the second person ingestion is failing

54. A meal of bread (starch), egg (protein) and butter (fat) is eaten. In which order are the chemical breakdowns of these three FIRST begun as the meal moves down the gut?

- (A) Protein first (in the mouth), then fat (in the stomach), then starch (in the small intestine)
- (B) Starch first (in the mouth), then protein (in the stomach), then fat (in the small intestine)
- (C) Fat first (in the mouth), then starch (in the stomach), then protein (in the small intestine)
- (D) All three begin their breakdown together in the stomach

55. A child loses the use of every canine but keeps all other teeth. A flesh-eating diet would suffer because canines do what, and which kept tooth type cannot substitute for that exact job?

- (A) Canines grind plant food; the incisors cut and so cannot grind in their place
- (B) Canines cut food at the front; the molars tear and so cannot cut in their place
- (C) Canines absorb nutrients in the mouth; no other tooth can absorb in their place
- (D) Canines tear flesh; the flat-topped molars and premolars grind rather than tear, so they cannot substitute for tearing

56. A scientist describes three organs only by action: Organ X churns food and mixes it with acid; Organ Y completes digestion and absorbs most nutrients; Organ Z recovers water from the leftover. Place them in the order food actually meets them.

- (A) Y (small intestine) first, then X (stomach), then Z (large intestine)
- (B) X (stomach) first, then Z (large intestine), then Y (small intestine)

- (C) Z (large intestine) first, then Y (small intestine), then X (stomach)
- (D) X (stomach) first, then Y (small intestine), then Z (large intestine)

57. An adult human has 32 teeth made of incisors, canines, premolars and molars. If exactly one quarter of all the teeth are the pointed-and-front cutting-and-tearing kinds combined, how many teeth are grinders, and does this match the standard set?

- (A) 8 are cutting-and-tearing and 24 are grinders, which exactly matches the standard set
- (B) 24 are cutting-and-tearing and 8 are grinders, matching the standard set
- (C) 8 teeth are cutting-and-tearing, so 24 would be grinders by this quarter-rule, which does NOT match the standard set of 12 cutting-and-tearing and 20 grinders
- (D) 16 are cutting-and-tearing and 16 are grinders, matching the standard set

58. A biologist states: 'In humans the place where food is digested is kept separate from the cells that finally use the nutrients.' Which pairing of locations supports this?

- (A) Digestion occurs inside the body cells, and the nutrients are used in the gut cavity
- (B) Digestion occurs in the blood, and the nutrients are used in the gut cavity
- (C) Both digestion and the using of nutrients happen only in the mouth
- (D) Digestion occurs in the cavity of the gut, and the nutrients are later used inside the body's cells, which are reached through the blood

59. A python swallows a whole egg, a butterfly sips nectar, and a cow grinds grass with broad teeth. A textbook places all three under one heading naming a single step, 'Tools for ____, shaped by the food eaten.' What completes the heading, and why are grinding teeth included with sipping and swallowing?

- (A) Digestion: all three tools chemically break the food down as they handle it
- (B) Absorption: all three tools pass nutrients into the body fluids at the feeding point
- (C) Assimilation: all three tools let the body use the food as it is gathered
- (D) Ingestion: all three are different tools for taking food in, and grinding teeth are part of how the cow takes its food in by preparing it for swallowing

60. Putting the journey together for a human: name the place where protein digestion BEGINS, the place where fat digestion is mainly COMPLETED, and the place where water is chiefly RECOVERED, in that order.

- (A) Stomach; small intestine; large intestine
- (B) Mouth; stomach; small intestine
- (C) Small intestine; mouth; large intestine
- (D) Large intestine; stomach; mouth

Solutions — Science (Class 7), Chapter 2:

Nutrition in Animals — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	A	21	C	31	D	41	B	51	C
2	B	12	C	22	B	32	A	42	C	52	A
3	B	13	D	23	D	33	A	43	D	53	C
4	B	14	B	24	A	34	A	44	C	54	B
5	A	15	A	25	C	35	D	45	A	55	D
6	D	16	A	26	B	36	B	46	B	56	D
7	B	17	B	27	A	37	C	47	A	57	C
8	D	18	B	28	C	38	D	48	C	58	D
9	D	19	A	29	C	39	A	49	C	59	D
10	B	20	C	30	D	40	C	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Taking the bite in is ingestion; the enzyme cutting the molecule into smaller pieces is digestion; those pieces crossing the gut wall into the transport fluid is absorption; a cell finally using them to build itself is assimilation. The option that puts assimilation before absorption is impossible, because nutrients cannot be used by a cell until they have first entered the transport fluid that reaches it. The list starting with digestion before ingestion is wrong because nothing can be broken down before it is taken in. The list placing absorption before digestion fails because a large molecule cannot cross the gut wall until it has been cut into small pieces.

2. (B) Molecules already present in the blood are the proof: their arrival there means food was taken in, broken down, and carried across the gut wall, so the first three steps worked. The wasting-away despite their presence shows the cells are not putting them to use, which is the assimilation step. The claim that they never reached the blood contradicts the stated fact that they are circulating, so absorption did happen. Ingestion plainly occurred since

food was taken in. Egestion concerns undigested residue leaving the body, not molecules already absorbed into blood, so it is unrelated to this failure.

3. (B) Egestion is specifically the removal of undigested food residue, which is exactly what Exit 1 carries. Exit 2 removes a gas produced inside cells during energy release, and any waste generated inside cells leaving the body is excretion. The label that calls Exit 1 excretion because it contains dead cells confuses the source: those dead cells travelled out mixed in the food residue, not as a chemical waste made for removal. The view that both are egestion ignores that egestion is tied to the food canal's leftovers, and the view that both are excretion ignores that fibre was never inside a cell.

4. (B) All three clips show food being taken into the body, which is ingestion; wrapping, swallowing whole, and sipping are merely different methods of the same step. They are not digestion, because none of the clips shows the chemical breakdown that happens after the food is inside. They are not absorption, because nutrients are not crossing into body fluids at the moment of feeding. They are not assimilation, because no cell is yet using anything; assimilation comes only after digestion and absorption.

5. (A) The external juice does not absorb anything into the fly; absorption means nutrients crossing into the body, and outside the body there is no body wall to cross. What the external juice does is start digestion, breaking solid food into a liquid that the fly then sucks in (ingestion) before any absorption occurs inside. The fly does not skip digestion; it merely starts some of it outside. It does not skip ingestion either, because sucking the liquid in IS ingestion.

6. (D) The point is about body plan: a single cell forms a temporary structure (the pseudopodium) that does double duty for movement and capture, whereas a many-organ body keeps a distinct fixed organ for each task. The extension does not perform digestion or absorption itself; those happen later inside a food vacuole, so the 'two chemical jobs at once' claim is wrong. It is not a permanent organ; its whole significance is that it is temporary. And it does not make bile; an Amoeba has no liver and the bile idea does not apply to single cells.

7. (B) Cutting teeth are the 8 incisors and tearing teeth are the 4 canines, giving 12; grinding teeth are the 8 premolars plus 12 molars, giving 20. Since 20 is greater than 12, grinding teeth outnumber cutting-and-tearing teeth, so the claim is false. The option that reverses the totals miscounts which teeth grind. There is no split of 16 and 16. And it is not the case that all 32 grind, since incisors and canines cut and tear rather than grind.

8. (D) The outermost shell is enamel, the hardest substance in the body; beneath it lies the softer dentine that makes up most of the tooth; and the central space carrying nerves is the pulp cavity, which is why pain begins only when the drill reaches it. Starting with dentine is wrong because dentine lies under the enamel, not outside it. Reaching the pulp

cavity before the dentine is impossible because the pulp is the innermost region. The fully reversed order puts the soft inner pulp outside the hard enamel, which cannot be.

9. (D) Pain in a tooth comes from the nerves, which lie in the pulp cavity at the centre. A cavity hurts only when it reaches those nerves, regardless of how wide or deep it looks: Cavity X, though deep in dentine, has not reached the pulp, while shallow Cavity Y has. Hardness has nothing to do with sensation, so 'dentine is hardest' is both false (enamel is hardest) and irrelevant. Nerves are in the pulp, not the enamel. And depth alone does not decide pain, so the two descriptions can indeed both be true.

10. (B) The chain begins with sugar sitting on the teeth, so removing that sugar by rinsing breaks the very first link, before bacteria can even start. The antacid idea is irrelevant because the acid that decays teeth is made by mouth bacteria, not risen from the stomach. Eating more sweets adds more sugar, feeding the bacteria rather than stopping them. Chewing tough food does not harden enamel against acid and acts, if at all, only at the last link rather than the earliest.

11. (A) Broad ridged teeth are suited to grinding plant matter and prominent pointed teeth to tearing flesh, so skull P fits a plant-eater and skull Q a flesh-eater; tooth form reflects diet. Tooth shape says nothing about body size, so the 'larger animal' inference is unwarranted. The two skulls are described with clearly different teeth, so they are not identical. And tooth form is not random; it reliably tracks the kind of food an animal eats.

12. (C) Premolars and molars do the grinding, so losing exactly those teeth makes grinding fibrous food into pulp the hardest task. Biting an apple uses the incisors, which were kept, so it stays possible. Tearing meat uses the canines, which were also kept. Swallowing liquids needs no teeth at all, so it is unaffected. Only the grinding function was removed.

13. (D) Only the starchy cracker turns sweet because the salivary enzyme breaks starch into sugar; the cheese, being fat and protein, has nothing the salivary enzyme can act on, so it stays unsweet. Saliva does not digest every food equally, so the cheese would not secretly sweeten. The enzyme does not act on protein, which is why neither would the cheese sweeten by that route. And it does not act on fat, so the fatty cheese cannot taste sweeter than the cracker.

14. (B) Both tubes were warm and both contained saliva, yet only the tube with fresh saliva made sugar; the one difference was that Tube 2's saliva had its enzyme destroyed by boiling, so the enzyme is what produced the sugar. Warmth alone cannot be the cause, since both tubes were equally warm. Acid is not the agent, since boiling destroys enzymes but would not remove acid, yet Tube 2 failed. And boiled saliva is not stronger; it lost its active enzyme.

15. (A) Beyond starting starch digestion, saliva wets and softens dry food into a slippery lump that can slide down the food pipe; without this, a dry biscuit could not be swallowed smoothly. Saliva does not absorb sugar into the blood; absorption happens far downstream at the small intestine, not the mouth. Saliva does not make hydrochloric acid; that belongs to the stomach. And cooling is not a recognised role of saliva in swallowing.

16. (A) Upside-down, gravity points away from the stomach, so if anything it should pull the water out of the mouth; the fact that the water still arrives proves the food pipe's muscular waves (peristalsis) push it there. Saying gravity helps ignores that gravity is working against the water in this position. The food pipe is not a straight vertical drop, and even if it were, position clearly matters when inverted. And the food pipe does not absorb the water into the blood; the water genuinely travels to the stomach.

17. (B) The food pipe moves food by peristaltic muscle waves; freeze those and swallowed food cannot be carried to the stomach, so it gets stuck there. The stomach still works, so its churning is unaffected and is not the direct problem. Saliva still works, so mouth digestion is unchanged. The food pipe is not where water is absorbed, so over-drying there is not the mechanism; the failure is simply transport.

18. (B) The wave-pushing tube passed in seconds is the oesophagus; the churning bag holding food a few hours is the stomach; the long coiled tube that retains food many hours for digestion and absorption is the small intestine, giving shortest-to-longest as oesophagus, stomach, small intestine. The stomach is not a seconds-long passage; it churns for hours. The small intestine, not the stomach, is the long-stay coiled tube, so the orders that swap them are wrong.

19. (A) Mucus does not digest anything; its job is to coat and shield the stomach wall, so calling it the protein-digester is the wrong assignment, with pepsin being the actual protein enzyme. Saying the acid kills germs is correct. Saying pepsin begins protein digestion is correct. Saying the acid creates the acidic conditions pepsin needs is also correct. Only the mucus statement is false.

20. (C) The acid does two jobs: it creates the acidic conditions pepsin needs and it kills germs in food. With little acid, pepsin works poorly so protein digestion in the stomach suffers, and germs are less well killed. Starch digestion does not depend on stomach acid, and bile's path to the small intestine is unrelated to acid. Fat emulsification depends on bile, not acid, and water absorption is the large intestine's separate job. Saliva and peristalsis are not produced by stomach acid.

21. (C) The wall really is made of protein, so the 'no protein' claim is simply false; what saves it is the mucus layer coating the inner surface, which keeps the acid and pepsin from reaching the living tissue. Bile is made in the liver and acts in the small intestine, not as a stomach coating. The wall is not bone; it is muscular protein tissue. The protection is the mucus barrier.

22. (B) The intestinal enzymes need non-acidic conditions, so the strongly acidic paste from the stomach must have its acidity reduced once it reaches the small intestine for those enzymes to act. Making it more acidic would suit only pepsin, not the intestinal enzymes. Keeping it stomach-acidic would let pepsin work but block the intestinal enzymes. Drying it out is not how digestion proceeds; the paste stays moist.

23. (D) The human stomach's semi-liquid paste is chyme, and the regurgitated grass a cow re-chews is cud, so the student swapped them. They are not both chyme; chyme is specifically the human stomach's output. They are not both cud; cud is the ruminant's regurgitated grass. And a cow does not re-chew bile, which is a fat-emulsifying fluid, not a chewable mass.

24. (A) The finger-like projections (villi) multiply the inner surface area, so much more of the gut wall is in contact with the food and far more absorption occurs in the same length. They do not speed food up; both guts share equal transit time anyway. They do not change acidity; absorption does not work by dissolving nutrients in acid. And villi do not make bile; bile comes from the liver.

25. (C) Flattening the villi destroys the large absorbing surface, so even well-digested nutrients are not taken up, and the person becomes undernourished and loses weight while still eating. The mouth cannot compensate; it only starts starch digestion and does no absorption. Harder churning by the stomach cannot replace the lost absorbing surface. And bile emulsifies fat; it does not absorb nutrients, so making more cannot fix lost villi.

26. (B) Glucose crossing the villus wall into the blood is absorption, and the muscle cell putting that glucose to use is assimilation, so the order is absorption then assimilation. Digestion is not happening at the villus; the glucose was already made before reaching it. Absorption cannot happen at the muscle, since the glucose entered the blood back at the villus. And egestion concerns undigested residue, not glucose in a muscle.

27. (A) Simple molecules show up at the villi because that is where they are absorbed, but they were produced earlier by enzymes acting in the gut cavity; the villi absorb, they do not digest. The villi are not the main digesting organs; digestion happens in the cavity. They do not store the molecules; the molecules pass into the blood. And they do not churn like a stomach; absorption, not churning, is their job.

28. (C) Breaking a fat globule into droplets is a preparation step within digestion, helping enzymes act; the amino acid crossing the wall into the blood is absorption. Both are not absorption: emulsifying fat does not cross any wall. Both are not digestion: crossing into blood is by definition absorption. And the labels are not reversed, since emulsification stays in the cavity while crossing the wall is absorption.

29. (C) Organ length does not decide which job an organ does; the small intestine specialises in absorbing nutrients, and recovering water from the leftover residue is

specifically the large intestine's role. So the small intestine does not also recover the water. The job split is not merely a matter of importance; it is a real division of labour. And the two do not share water recovery equally; it is chiefly the large intestine's task.

30. (D) Water left in the residue makes stools loose and watery, so too little reabsorption gives watery stools; reabsorbing too much dries the residue into hard stools (constipation). The reversed pairing is wrong because more water means wetter stools, not drier. Vomiting acid and fat digestion are unrelated to large-intestine water recovery. And the large intestine does handle water; recovering it is its main job.

31. (D) Being indigestible, the bead can only be taken in (ingestion) and later leave with the undigested residue (egestion); no enzyme breaks it, so digestion, absorption and assimilation never apply to it. Stomach acid will not turn an inert bead into nutrients. It cannot be assimilated into bone, since assimilation needs absorbed nutrients. And it is not excretion, because excretion removes waste made inside cells, not a swallowed object passing through the canal.

32. (A) Bile helps without being an enzyme: in the small intestine it physically breaks large fat globules into tiny droplets, vastly increasing the surface on which fat-digesting enzymes can work, so it is far from useless. Bile does not chemically cut fat into amino acids; amino acids come from protein anyway, and bile carries no enzyme. And bile does not convert fat into starch; that conversion is not a real process.

33. (A) Because only fatty meals are troublesome, the lost fluid must be one that mainly helps fat: that fluid is bile, which the gall bladder stored, and bile assists fat digestion by emulsifying it. The gall bladder does not make pepsin; pepsin is a stomach protein enzyme. It does not store amylase; if it had, starchy meals would suffer, which they do not. And it does not make stomach acid; the selective effect on fats rules out an effect on all foods.

34. (A) Bile's job is to emulsify fat; without it, fat remains in large globules that enzymes can hardly act on, so fat digestion suffers most, while starch and protein still have their working saliva, gastric and pancreatic enzymes. Bile is not a starch enzyme, so starch is not the worst hit. Bile does not activate the protein enzymes, so protein is not the worst hit. And bile is not needed for every food, so the three are not equally hampered.

35. (D) Saliva's enzyme chemically begins starch digestion, gastric pepsin chemically begins protein digestion, and bile only physically emulsifies fat into droplets without itself breaking the chemical bonds. The first wrong option misassigns every pairing. The second has saliva emulsifying fat and pepsin on starch, both false. And the third is impossible, since the three act on different foods, not all on starch.

36. (B) Pancreatic juice carries enzymes for carbohydrates, proteins and fats acting in the small intestine, so blocking it impairs all three there at once. Mouth starch digestion uses salivary enzyme, not pancreatic, so it continues. Stomach protein digestion uses pepsin,

made by the stomach, not the pancreas. And the pancreas does not absorb water; that is the large intestine's job, unaffected here.

37. (C) The pancreatic enzymes act only in non-acidic surroundings, so the strongly acidic chyme must be neutralised in the small intestine before they can work. They do not need stronger acid than pepsin; the opposite is true. Neutralising acid does not itself digest food; it only sets the right conditions for the enzymes. And the intestine does hold moist contents; dryness is not required.

38. (D) The largest chamber where microbes break down cellulose is the rumen, and the chamber doing acid-and-enzyme digestion like a human stomach is the abomasum. The swap that puts microbes in the abomasum and acid in the rumen reverses their real roles. The reticulum is not the acid-and-enzyme chamber, and the omasum is not where microbial cellulose breakdown begins; the rumen is.

39. (A) Grass is rich in cellulose, which neither animal's own enzymes can break; the cow manages only because microbes in its rumen plus repeated cud-chewing break the cellulose down, a system the person lacks entirely. A bigger stomach alone would not help, since the limiting issue is cellulose-digesting microbes, not capacity. The person does have a stomach. And grass certainly needs digestion; eating more of an undigestible food does not nourish.

40. (C) Chewing the cud only grinds the grass finer; the actual breakdown of cellulose is done by microbes in the rumen, not by chewing in the mouth. The mouth has no cellulose-digesting power, so it cannot fully digest cellulose. The food pipe merely transports the cud and does not break cellulose. And the cellulose is not stored forever; it is digested in the rumen with help from the finer grinding.

41. (B) Grinding the cud finer exposes more grass surface, letting rumen microbes and enzymes act on more of the cellulose and free more nutrients, so more energy becomes available. The mouth does not turn cellulose into sugar for absorption there; cellulose breakdown happens in the rumen. Chewing does not release energy by heat. And finer grinding does not convert cellulose into protein; that conversion does not occur.

42. (C) Inside an Amoeba the food is trapped in a food vacuole, and the undigested leftover leaves when the vacuole reaches the cell surface and opens, which is egestion. A gall bladder and a rumen are organs of larger animals, not parts of a single cell. A villus belongs to the human small intestine. And throwing out leftovers is egestion, not absorption (taking in) or assimilation (using nutrients).

43. (D) The food vacuole is an enclosed pocket where enzymes meet food, and the human parallel is the enclosed cavity of the stomach and intestine, where digestive enzymes act on food. Human digestive enzymes do not act in the blood; the blood carries already-absorbed

nutrients. They do not act inside muscle cells, where nutrients are merely used. And they do not act on the skin; food is digested inside the gut, not outside the body.

44. (C) Both organisms carry out all five steps; the only difference is that the Amoeba does them inside one cell while the human uses many specialised organs. The Amoeba does digest (in its food vacuole), so the 'never digests' claim is false. The Amoeba does egest leftovers when the vacuole reaches the surface, so it completes all five. And the human does assimilate, using absorbed nutrients in its cells, so 'never assimilates' is false.

45. (A) Starch digestion begins in the mouth with the salivary enzyme, while fat is mainly emulsified by bile and digested by enzymes in the small intestine. Neither begins in the stomach for these two foods: the stomach handles protein, not starch or fat emulsification. Starch does not first begin in the small intestine, and fat is not digested in the mouth. The large intestine recovers water and does not digest starch or fat.

46. (B) Protein digestion begins in the stomach with pepsin, is completed in the small intestine by its enzymes, and the resulting building blocks are absorbed there at the villi. Protein digestion does not begin in the mouth, since saliva acts on starch, not protein. It is not completed in the stomach; the small intestine finishes it. And absorption of the building blocks happens in the small intestine, not the mouth, stomach or large intestine.

47. (A) Dissolved sugar is mainly absorbed across the villi of the small intestine, while most water is recovered later in the large intestine. The stomach is mainly for churning and starting protein digestion, not for absorbing sugar and water. The pairing that swaps the two organs is wrong because sugar absorption is the small intestine's job and water recovery the large intestine's. And the mouth does not absorb either into the blood.

48. (C) At the instant the milk enters the mouth it is only being taken in, so ingestion is occurring and no step has yet been completed. Digestion has not been completed, and absorption cannot occur this early. Two steps cannot be done at the moment of entry, so assimilation is not occurring. And four steps cannot be finished before the food is even fully in, so egestion is not the only step left.

49. (C) The only error is putting assimilation before absorption: a cell cannot use a nutrient until that nutrient has crossed into the blood, so absorption must precede assimilation. Egestion correctly comes last, not before digestion, because it removes the final undigested residue. Ingestion must be first, not last, since food has to be taken in before anything else. And the two orders are not equivalent, because the order of steps is fixed by what each step needs from the one before.

50. (B) The largest gland making bile is the liver; the small sac storing bile between meals is the gall bladder; and the gland whose juice digests carbohydrates, proteins and fats is the pancreas, giving liver, gall bladder, pancreas. The orderings that move the gall bladder

or pancreas out of place mismatch the descriptions, and putting the pancreas first wrongly assigns bile-making to the gland that actually makes the all-food-type juice.

51. (C) Without peristalsis the food pipe cannot push swallowed food to the stomach, so the food lodges in the pipe; that is the first medicine's effect. The loss of mouth sweetness on chewing is the SECOND medicine's effect (the salivary enzyme), not the peristalsis one. The stomach still works, so churning is unaffected. And the large intestine still recovers water, so that is unrelated to the food pipe.

52. (A) Swallowing whole and engulfing are two ways of taking food in, so they differ in ingestion, not digestion; digestion is the chemical breakdown that comes later and is not what these scenes show. They are not absorption, which is nutrients crossing into body fluids, not yet shown. They are not two kinds of digestion, since neither scene shows chemical breakdown. And assimilation is the later cell use of nutrients, also not depicted.

53. (C) The first person absorbs glucose into the blood but the cells cannot use it, so the failing step is assimilation; the second person gets nothing into the blood, so the failing step is absorption. The first is not absorption, because glucose did reach the blood. Neither failure is digestion, since food is broken down in both descriptions. And egestion and ingestion are not the failing steps, since the problems lie with using and with crossing into blood.

54. (B) Starch digestion begins first, in the mouth, by the salivary enzyme; protein digestion begins next, in the stomach, by pepsin; fat digestion begins last, in the small intestine, with bile and enzymes. Protein does not begin in the mouth, and fat does not begin in the stomach, so the first wrong option is misordered. Fat does not begin in the mouth, ruling out the second. And the three do not all begin together in the stomach; starch starts earlier in the mouth.

55. (D) Canines are the pointed teeth that tear flesh, and the molars and premolars are flat grinders, so they cannot take over the tearing job. Canines do not grind; that is the molars' role, so the grinding claim is wrong. Canines tear rather than cut; cutting is the incisors' job. And teeth do not absorb nutrients at all, so the absorption claim is false.

56. (D) The acid-churning organ is the stomach (X), met first; next is the small intestine (Y), which completes digestion and absorbs most nutrients; last is the large intestine (Z), which recovers water, so the order is X, Y, Z. Food cannot reach the small intestine before the stomach, ruling out Y first. The large intestine comes after, not before, the small intestine, ruling out the orders that put Z early.

57. (C) One quarter of 32 is 8, so the quarter-rule would give 8 cutting-and-tearing teeth and 24 grinders; but the real adult set has 8 incisors plus 4 canines, that is 12 cutting-and-tearing, and 20 grinders, so the quarter-rule does NOT match. Saying 24 grinders matches

is wrong because the true count is 20. Reversing to 24 cutting teeth is far from the real 12. And a 16-16 split matches neither the rule nor the real set.

58. (D) Digestion takes place in the enclosed cavity of the gut, while the cells that use the nutrients lie elsewhere in the body and receive them via the blood, so the two locations are indeed separate. Digestion is not inside the cells, and using nutrients is not done in the gut cavity, so that pairing is reversed. Digestion does not happen in the blood. And neither digestion nor nutrient use is confined to the mouth.

59. (D) Swallowing whole, sipping, and grinding-before-swallowing are all ways of taking food into the body, so the heading is ingestion, and the cow's grinding teeth are part of its ingestion by readying the grass. They are not tools of digestion, since chewing prepares food but the chemical breakdown comes later. They are not absorption tools, since nothing crosses into body fluids at the mouth. And they are not assimilation tools, which would be cells using nutrients.

60. (A) Protein digestion begins in the stomach with pepsin, fat digestion is mainly completed in the small intestine with bile and enzymes, and water is chiefly recovered in the large intestine, giving stomach, small intestine, large intestine. Protein does not begin in the mouth, where saliva acts on starch. Fat is not completed in the stomach or digested in the mouth. And water is recovered in the large intestine, not the mouth.